Problem Set 1:
Computer Networks and the Internet: Introduction

1 Essay Questions

1. Describe the most popular wireless Internet access technologies today. Compare and contrast them.

2. What advantage does a circuit-switched network have over a packet-switched network? What advantages does TDM have over FDM in a circuit-switched network?

3. Consider sending a packet from a source host to a destination host over a fixed route. List the delay components in the end-to-end delay. Which of these delays are constant and which are variable?

4. What are the five layers in the Internet protocol stack? What are the principal responsibilities of each of these layers?

2 Problems

1. Consider an application which transmits data at a steady rate (e.g., the sender generates an N bit unit of data every k time units, where k is small and fixed). Also, when such an application starts, it will stay on for a relatively long period of time. Would a packet-switched network or a circuit-switched network be more appropriate for this application? Why?

2. Suppose there is a 10 Mbps microwave link between a geostationary satellite and its base station on Earth. Every minute the satellite takes a digital photo and sends it to the base station. Assume a propagation speed of $2.4 * 10^8$ meters/sec.

   (a) What is the propagation delay of the link?

   (b) What is the bandwidth-delay product, R * d_{prop}?

   (c) Let x denote the size of the photo. What is the minimum value of x for the microwave link to be continuously transmitting?

3. Suppose Host A wants to send a large file to Host B. The path from Host A to Host B has three links, of rates $R_1 = 500$ kbps, $R_2 = 2$ Mbps, and $R_3 = 1$ Mbps.

   (a) Assuming no other traffic in the network, what is the throughput for the file transfer?

   (b) Suppose the file is 4 million bytes. Dividing the file size by the throughput, roughly how long will it take to transfer the file to Host B?

   (c) Repeat (a) and (b), but now with R_2 reduced to 100 kbps.

4. Suppose users share a 2 Mbps link. Also suppose each user transmits continuously at 1 Mbps when transmitting, but each user transmits only 20 percent of the time. (See the discussion of statistical multiplexing in Section 1.3.)

   Answer the following questions:

   (a) When circuit switching is used, how many users can be supported?

   (b) For the remainder of this problem, suppose packet switching is used. Why will there be essentially no queuing delay before the link if two or fewer users transmit at the same time? Why will there be a queuing delay if three users transmit at the same time?

   (c) Find the probability that a given user is transmitting.

   (d) Suppose now there are three users. Find the probability that at any given time, all three users are transmitting simultaneously.

3 Notes

- It's encouraged that you take a look at the Homework Problems for chapter 1, from the kurose/Ross reference.

Good Luck